

## CURRICULUM VITAE

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|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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| Date and city of birth: | 1 May 1970      Amsterdam                                                                                                                                                                                                              |
| Education:              | 1982 - 1989      Lorentz Lyceum VWO, Eindhoven<br>1989 - 1990      TU-Delft mechanical engineering<br>1990 - 1994      HTS-Haarlem mechanical engineering<br>1994 - 1997      TU-Delft mechanical engineering transportation technique |
| Certificates:           | 1989      Atheneum-B<br>1994      HTS mechanical engineering 'Cum Laude'<br>1997      TU mechanical engineering                                                                                                                        |

### Relevant working experience:

|             |                                                                             |
|-------------|-----------------------------------------------------------------------------|
| 1998 – 2002 | Itrec – mechanical design engineer                                          |
| 2002 – 2004 | Itrec – lead engineer                                                       |
| 2004 – 2006 | Itrec – senior mechanical engineer                                          |
| 2006 – 2010 | Bluewater Energy Services – senior mechanical engineer                      |
| 2010 -      | Minahassa Mechanical Engineering - owner; mechanical engineering consultant |

### Patents:

Tubular handling apparatus and a drilling rig WO2006038790

This patent describes a tubular handling apparatus for a drilling rig. Some sequential handling stages are combined in one single step. Design is simplified: lighter; less actuators and quicker operation.

Mooring assembly – application no 08160604.8 – 2422

The patent describes several solutions for FPSO main bearing replacement during operation.

### Design/fabrication projects:

#### Supply boat mooring buoy

For a 2500 tonne deadweight supply boat mooring buoy I am specifying the mechanical items

**Design proposal for a large FPSO turret – BP**

For the proposal of a large turret (>20m diameter) of an FPSO I modified the design in order to achieve full replacement of key mechanical items. Replacement of the mooring chains and the chain stoppers is possible without diver intervention. Furthermore it is possible to transfer the load of the main bearing onto a secondary bearing on the field in order to allow for bearing overhaul.

**LNG tandem offloading system**

Bluewater has developed a tandem offloading system for offshore LNG transfer. The system is based on utilization of a newly developed composite cryogenic hose. I made the mechanical design of the system.

**Large bearings for FPSO's – in house study**

I am involved in a study on the implications of further increase of FPSO main bearing diameters. Solutions for bearing replacement are being patented.

**FPSO double pivoting mooring chain stoppers – BP Skarv**

As part of a FEED study for BP I made the design for double pivoting chain stoppers for the FPSO intended for the SKARV field. A double pivoting mechanism was designed in order to reduce out-of-plane bending of the mooring chains, a phenomenon that in the recent past has led to several mooring chain failures. The chain stoppers were designed for 25 years uninterrupted service, with no maintenance possible. Peak load for each chain stopper is approximately 1400 tonne.

**Disconnectable turret mooring system – Bluewater FPSO Munin**

A feed study was carried out for Petrobras. The study was concerning a disconnectable turret mooring system for FPSO Munin. In case of a hurricane the FPSO could disconnect from the mooring and the risers and leave the field. Risers and mooring chains would remain connected to a buoy that - when disconnected – was to sink to a depth of approx. 25m. I made the design of the buoy pull in system that is to pull the buoy back into a receptacle in the turret.

**LOC 250 – several clients**

LOC 250 is a drilling rig built from modules that fit in standard ISO containers and can be installed in several days. I made the concept design of several modules and I invented the –patented- mechanism for the pipe loading machine which loads all tubulars (drillpipe, casing) into the tower.

**Workstation Seven Oceans and PLET line up tool – Subsea 7**

I made the basic design of the workstation that is placed on the pipe lay tower of Seven Oceans. The workstation has large openings on SB, on aft and on bottom side. The main purpose of the workstation is creating a sheltered working area for welding Pipe Line End Terminations to the pipe in the firing line. For this the PLETs are loaded from the side placed on dedicated pallets.

**Line up tool for gimbaling J-lay tower – Stolt Offshore (now: Acergy)**

I made the basic design for the line up tool for the new J-lay tower for Stolt offshore. This is a very special tower, because it gimbals about two axes, which enables the vessel to remain working in head seas direction, while laying pipe in cross direction, in backward or forward direction or any direction inbetween.

**Test/transport frame and hoisting provisions for line up tool for gimbaling J-lay tower – Jumbo Shipping**

I designed the test/transport frame for abovementioned J-lay tower. The frame was required to spread the load of the tower + test weights onto the quayside. After testing the frame served as seafastening for the tower on the heavy lift vessel for transport to Gabon. I also designed the lifting points on the tower; both in co-operation with Jumbo.

**15" pipe aligner and straightener system for pipe with piggy back pipe – Torch offshore (now Helix Express)**

My design team designed the aligner and straightener system for the reel lay vessel from Torch. This system straightens the rigid pipe before it is lowered to the seabed.

**160t hang off module – Torch offshore**

My design team designed the hang off module for a reel lay pipe laying vessel for Torch offshore. This hang off module skids over a rail on the tower in order to work in two firing lines. It has a side opening over the entire length that enables side loading around the sea pipe. The pipe is caught by means of friction pads that are pressed to the pipe with hydraulic cylinders.

#### **400t heave compensation system for offshore crane – Sealion**

For a 400t offshore crane built for Sealion, my design team designed a passive and active heave compensation system. The system consists of a large hydraulic cylinder mounted on the bottom of the traction winch, combined with a medium separator and a nitrogen tank and a hydraulic pump for the active heave control. I also supervised making of the shopdrawings and gave support during commissioning.

#### **Line up tool for J-lay system – Heerema**

This system lines up hex pipes (72m length) to the sea pipe that hangs in the hang off module, in order to enable welding the sections together. In some configurations of the J-lay tower, the transverse loading occurring during line up may be up to 100t for each actuator set.

#### **Riser handling crane - Petrodrill**

The Riser Handling is a dedicated gantry crane that can handle marine risers on a semisubmersible drillship. The risers are picked up from their storage position in the leg of the semi, and laid horizontally onto the crane. Then the crane travels to the centerline of the ship where it hands over the risers to the derrick. For this crane I designed several subsystems, I supervised production of shopdrawings and I attended the installation on a rig at Daewoo shipyard in Korea.

### **Design validation testing projects:**

#### **LNG hose qualification programme**

For the qualification of the Composite Cryogenic Hose some 25 qualification tests are being carried out. I am defining the required test protocols and supervising the tests. I have made the concept design of the dedicated fatigue test rig and supervised the detail design. Currently pressure tests have been carried out at the fabrication site of Dantec in Wirral, UK. Also flow tests have been carried out at Deltares in Delft, The Netherlands. I assisted in qualification of the key suppliers.

#### **Large scale sliding bearing testing at Labo Soete, University of Gent, Belgium**

For the chain stoppers of the BP-Skarv FPSO mentioned above a qualification programme was set up for the sliding bearings. The purpose was to select the best bearing material and to obtain friction and wear data. For Bluewater I supervised the first two phases (out of 3) of this project.

#### **Polyurethane tensioner squeeze pads for Ormen Lange (Skandi Neptune) – Subsea 7**

On request of Subsea 7 Huisman has fitted out tensioner squeeze pads for umbilicals with PU coating. The first batches did not function satisfactory. Together with Subsea 7 and the suppliers an improvement programme was carried out, including re-design and testing at Bodycote laboratory in Daventry, UK. For Huisman I took care of the technical part of this project.

#### **Fatigue test rig for hoisting wire – Huisman**

Huisman has delivered the Multi Purpose Tower on the Q4000, now owned by Helix Energy Solutions. In order to validate the calculated fatigue life of the hoisting wire I have designed a test rig. The rig is capable of simulating bending, counter bending and side lead of the wire. In this test rig the wire was successfully tested.

#### **Large scale bearing testing at Orkot, Rotherham, UK**

For the main bearings of the gimballing J-lay tower for Stolt (Acergy) a test programme was carried out at Orkot in order to validate bearing integrity under dynamic and thermal load. For Huisman I supervised the tests.

### **Repair/overhaul projects**

#### **Utility swivel overhaul FPSO Triton**

Together with a colleague I overhauled the utility swivels of FPSO Triton. The swivel stack was disassembled. Then one module was changed out; two modules were overhauled on board. Finally the stack was built up again and pressure tested.

#### **Utility swivel overhaul FPSO Bleo Holm**

I assisted in the overhaul of the utility swivels of FPSO Bleo Holm, an operation similar to overhaul of the Triton utility swivels.

**Crude oil offloading buoy main bearing exchange, Pointe Noire, Congo**

This job consisted of: jacking up of the buoy turntable, lifting off the bearing, lifting on the new bearing, alignment of the turntable and tightening of the bearing. All work was carried out with a local crew.

**Repair of damaged crane, Daewoo Shipyard, South Korea**

For this project I investigated the damage on the –just installed- Riser Handling Crane caused by a collapsing harbour crane. In a later stage I returned in order to start up the repairs programme.